

VINAY B R

AI/ML Developer | Deep Learning | Model Deployment

✉ <mailto:vinaybr0718@gmail.com>

🌐 <https://linkedin.com/in/vinay-b-r>

☎ [tel:+91 80887 57674](tel:+918088757674)

📍 <https://github.com/VinayBR03>

📍 Bengaluru, India

PROFILE SUMMARY

Entry-level AI/ML engineer with hands-on experience building and deploying machine learning solutions across NLP and reinforcement learning domains. Proficient in Python, deep learning frameworks, and REST-based model deployment pipelines. Passionate about developing AI systems that address real-world challenges, with a focus on IoT-integrated and edge-optimized solutions.

TECHNICAL SKILLS

- **Programming:** Python, Java, SQL, C++ (Arduino/ESP32), Solidity
- **Machine Learning & Deep Learning:** Scikit-learn, TensorFlow, PyTorch, NumPy, Pandas
- **NLP:** Hugging Face Transformers, BERT, NLTK
- **Backend:** FastAPI, Node.js, Flask, MySQL, PostgreSQL, Socket.io, MQTT, REST APIs
- **Frontend:** React, Vite, React Native, Expo, Tailwind CSS, HTML/CSS
- **IoT:** ESP32, LoRa, PlatformIO, Sensor Integration (MAX30102)
- **Tools:** Git, GitHub, Docker, Google Colab, VS Code

PROJECTS

SentinelTour - Smart Tourist Safety System FastAPI, Geofencing, MQTT, TinyML, Blockchain

- Engineered **ESP32** and **MQTT** monitoring over **LoRa** mesh networks for reliable tracking in low-connectivity zones.
- Deployed **TinyML** for edge fall detection and streaming AI to identify erratic or high-risk tourist behavior patterns.
- Developed an **Async FastAPI** and **PostgreSQL** backend for real-time sensor data orchestration and geofencing.
- Integrated **Solidity-based smart contracts** for immutable SOS logging; developed a **React Native** mobile app for traveler-side safety and a **React** web dashboard for authority-side incident orchestration and real-time response.

Face Liveness Detection System

Python, PyTorch, ONNX, TFLite, Flask

- Built an end-to-end **anti-spoofing system** to distinguish live human faces from printed photos, video replays, and masks.
- Designed a multi-modal architecture combining **MobileNetV3** with **LSTM-based temporal modelling** and sensor fusion.
- Implemented a **PyTorch to TFLite** conversion pipeline; **INT8 quantised** model achieves 95-98% accuracy at only 1.5-2 MB.
- Deployed a **Flask REST API** and webcam pipeline for real-time inference on edge devices.

ICU Decision Support System

Python, Q-Learning, Flask

- Built a **reinforcement learning-based system** to recommend optimal ICU treatment strategies from patient vital signs using **Q-learning**.
- Modelled the clinical environment with defined **state spaces** and **reward structures** to reflect real-world ICU decision scenarios.
- Delivered a **Flask web interface** with interactive input forms and result visualisation for outcome validation.

Domestic Violence Risk Detector

Python, BERT, Hugging Face, Flask

- Developed an **NLP-based risk assessment tool** to evaluate domestic violence risk from text and audio inputs.
- **Fine-tuned BERT** for **multi-class risk classification** and integrated **speech-to-text** preprocessing for audio support.
- Exposed the model via a **Flask API** delivering probability-based risk scores for real-time support service usage.

EDUCATION

Presidency University, Bengaluru

B.Tech in Computer Science & Engineering (AI & ML Specialisation)

📅 2022 - 2026

KLE Independent PU College, Bengaluru

Pre-University (PCMC)

📅 2020 - 2022

CERTIFICATIONS

- **Build Generative AI Apps and Solutions with No-Code Tools** - Infosys Springboard
- **React (Basic)** - HackerRank
- **Introduction to Supervised and Unsupervised ML** - Simplilearn
- **Data Visualization Techniques** - Infosys Springboard

ACHIEVEMENTS & ACTIVITIES

- Participated in the **Google Solution Challenge**.
- Completed the **Deloitte Data Analytics Job Simulation** (Forage).